

Task 1: To Parse Combined Redirection, Duplicate File Descriptors, Merge Streams, Verify Ordering, Handle Edge Cases, Test Integrated Outputs

```
GNU nano 7.2 task1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>

int main() {
    int fd;

    fd = open("combined.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1) {
        perror("open");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    dup2(STDOUT_FILENO, STDERR_FILENO);

    close(fd);

    printf("Message from stdout\n");
    fprintf(stderr, "Message from stderr\n");
    printf("Second stdout message\n");
    fprintf(stderr, "Second stderr message\n");

    return 0;
}
```

```
shreyas@Grigori:~$ mkdir -p ~/combined_redirection
cd ~/combined_redirection
shreyas@Grigori:~/combined_redirection$ nano task1.c
shreyas@Grigori:~/combined_redirection$ gcc task1.c -o task1
shreyas@Grigori:~/combined_redirection$ ./task1
shreyas@Grigori:~/combined_redirection$ cat combined.txt
Message from stderr
Second stderr message
Message from stdout
Second stdout message
shreyas@Grigori:~/combined_redirection$ ./task1 > output.txt 2>&1
shreyas@Grigori:~/combined_redirection$ cat output.txt
shreyas@Grigori:~/combined_redirection$ nl -ba combined.txt
  1 Message from stderr
  2 Second stderr message
  3 Message from stdout
  4 Second stdout message
shreyas@Grigori:~/combined_redirection$ ls -l /proc/$$/fd
total 0
lrwx----- 1 shreyas shreyas 64 Sep 27 14:17 0 -> /dev/pts/2
lrwx----- 1 shreyas shreyas 64 Sep 27 14:17 1 -> /dev/pts/2
lrwx----- 1 shreyas shreyas 64 Sep 27 14:17 2 -> /dev/pts/2
lrwx----- 1 shreyas shreyas 64 Sep 27 14:17 255 -> /dev/pts/2
lrwx----- 1 shreyas shreyas 64 Sep 27 14:17 7 -> /dev/pts/2
shreyas@Grigori:~/combined_redirection$ rm -f task1 combined.txt output.txt
shreyas@Grigori:~/combined_redirection$
```

Task 2: To Parse Complex Commands, Combine Pipes and Redirections, Resolve Execution Order, Validate Syntax, Generate Execution Plans, Test Scenarios

```
GNU nano 7.2 task2.
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <fcntl.h>

int main() {
    int pipefd[2];
    pid_t p1, p2;

    if (pipe(pipefd) == -1) {
        perror("pipe");
        return 1;
    }

    p1 = fork();

    if (p1 == -1) {
        perror("fork");
        return 1;
    }

    if (p1 == 0) {
        int fd = open("file1.txt", O_RDONLY);

        if (fd == -1) {
```

```

GNU nano 7.2 task2.c *
}

p2 = fork();

if (p2 == -1) {
    perror("fork");
    return 1;
}

if (p2 == 0) {
    int fd = open("result.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1) {
        perror("result.txt");
        exit(1);
    }

    dup2(pipefd[0], STDIN_FILENO);
    dup2(fd, STDOUT_FILENO);

    close(fd);
    close(pipefd[0]);
    close(pipefd[1]);

    execlp("uniq", "uniq", NULL);
}

shreyas@Grigori:~/complex_commands$ echo "apple" > file1.txt
echo "banana" >> file1.txt
echo "apple" >> file1.txt
echo "orange" >> file1.txt
echo "banana" >> file1.txt
shreyas@Grigori:~/complex_commands$ cat file1.txt
apple
banana
apple
orange
banana
shreyas@Grigori:~/complex_commands$ nano task2.c
shreyas@Grigori:~/complex_commands$ gcc task2.c -o task2
shreyas@Grigori:~/complex_commands$ ./task2
Execution completed successfully.
shreyas@Grigori:~/complex_commands$ cat result.txt
apple
banana
orange
shreyas@Grigori:~/complex_commands$ sort < file1.txt | uniq > test_result.txt
shreyas@Grigori:~/complex_commands$ cat test_result.txt
apple
banana
orange
shreyas@Grigori:~/complex_commands$ diff result.txt test_result.txt
shreyas@Grigori:~/complex_commands$ sort < missing.txt | uniq > error_test.txt
-bash: missing.txt: No such file or directory
shreyas@Grigori:~/complex_commands$ ls -l error_test.txt
-rw-r--r-- 1 shreyas shreyas 0 Sep 27 14:21 error_test.txt
shreyas@Grigori:~/complex_commands$ ls -l

```

```
orange
shreyas@Grigori:~/complex_commands$ sort < file1.txt | uniq > test_result.txt
shreyas@Grigori:~/complex_commands$ cat test_result.txt
apple
banana
orange
shreyas@Grigori:~/complex_commands$ diff result.txt test_result.txt
shreyas@Grigori:~/complex_commands$ sort < missing.txt | uniq > error_test.txt
-bash: missing.txt: No such file or directory
shreyas@Grigori:~/complex_commands$ ls -l error_test.txt
-rw-r--r-- 1 shreyas shreyas 0 Sep 27 14:21 error_test.txt
shreyas@Grigori:~/complex_commands$ ls -l
total 36
-rw-r--r-- 1 shreyas shreyas    0 Sep 27 14:21 error_test.txt
-rw-r--r-- 1 shreyas shreyas   33 Sep 27 14:20 file1.txt
-rw-r--r-- 1 shreyas shreyas   20 Sep 27 14:21 result.txt
-rwxr-xr-x 1 shreyas shreyas 16392 Sep 27 14:21 task2
-rw-r--r-- 1 shreyas shreyas  1373 Sep 27 14:21 task2.c
-rw-r--r-- 1 shreyas shreyas    20 Sep 27 14:21 test_result.txt
shreyas@Grigori:~/complex_commands$ rm -f task2 file1.txt result.txt result2.txt test_result.txt error_test.txt
shreyas@Grigori:~/complex_commands$ |
```